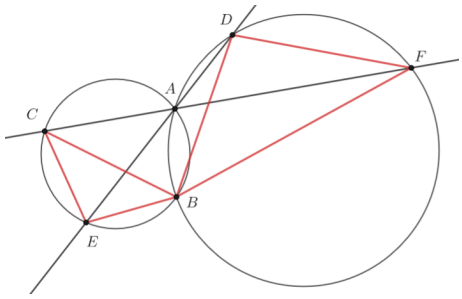
Working With Spiral Similarity

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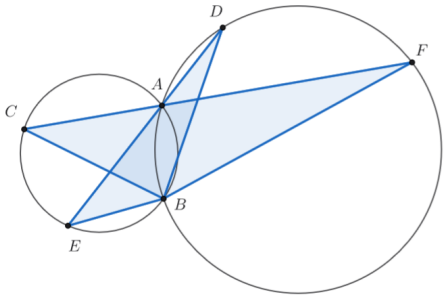
August 2026

1 Toolbox

Theorem 1: Let ω_1 and ω_2 be circles intersecting at A and B. Lines ℓ_1 and ℓ_2 pass through A, and intersect ω_1 at C and E and ω_2 at D and F. Then B is the centre of the spiral similarity sending CE to FD.



Theorem 2 (Switching the Order): If B is the centre of a spiral similarity sending CE to FD, it is also the centre of a spiral similarity sending ED to CF.



Idea (Helpful from problem 5 onwards.): Sometimes when we try to solve a problem with an important spiral similarity, it can help to add in the images of extra points under the spiral similarity.

2 Problems

Problem 1. Let C be a point on line AE, and construct triangles ABC and CDE on the same side of line AE. Let R be the intersection of AD and BE. Prove that RC bisects $\angle ARE$.

Problem 2. **(a)** (Brazil 2011) Let ABC be an acute triangle with orthocenter H and altitudes BD, CE. The circumcircle of $\triangle ADE$ cuts the circumcircle of $\triangle ABC$ at $Q \neq A$. Prove that the angle bisectors of $\angle BQC$ and $\angle BHC$ concur at a point on BC.

(b) Let ABC be an acute triangle with incenter I and incircle touchpoints D, E, F on BC, AC, AB respectively. The circumcircle of $\triangle AEF$ cuts the circumcircle of $\triangle ABC$ at $S \neq A$. Let M be the midpoint of arc BC on the circumcircle of $\triangle ABC$. Show that S, D, M are collinear.

Problem 3. (China 1992) Convex quadrilateral $ABCD$ is inscribed in circle ω with center O. Diagonals AC and BD meet at P . The circumcircles of triangles ABP and CDP meet at P and Q. Assume that points O, P, and Q are distinct. Prove that $\angle OQP = 90^\circ$.

Problem 4. (IMO Shortlist 2006) Let $ABCDE$ be a convex pentagon with $\angle BAC = \angle CAD = \angle DAE$ and $\angle CBA = \angle DCA = \angle EDA$. The diagonals BD and CE meet at P. Prove that line AP bisects CD.

Problem 5. (Russia 2014) Let ABC be a triangle with $BA > BC$ and circumcircle Γ. Let M, N be points on AB and BC with $AM = CN$. Let K be the intersection of $\overline{MN}$ and $\overline{AC}$. Denote by P the incenter of $\triangle AMK$ and Q the K-excenter of $\triangle CNK$. If R is the midpoint of arc ABC of Γ, prove that $RP = RQ$.

Problem 6. (EMO 2026 P7) Let ABC be a triangle and let M be the midpoint of segment BC. Let E and F be points on segments AC and AB, respectively, such that the circumcircle of triangle MEF is tangent to BC. The circumcircles of triangles AEF and ABC intersect at a point $P \neq A$. Let Q be a point on the circumcircle of triangle AEF such that AQ is perpendicular to BC.

Prove that PQ passes through the circumcenter of triangle MEF.

Problem 7. (USAMO 2026 P5) Let ABC be a triangle. Points D, E, F lie on sides BC, CA, AB, respectively, such that

$$\angle AFE = \angle BDF = \angle CED.$$

Let O_A, O_B, O_C be the circumcenters of triangles AFE, BDF, CED, respectively. Let M, N, O be the circumcenters of triangles ABC, DEF, $O_AO_BO_C$, respectively. Prove that $OM = ON$.

Problem 8. (Original) Let $\triangle ABC$ have incenter I and incircle ω. Let X be on AB and Y be on AC, both between the touchpoints of ω and A, so that $\overline{XY}$ is tangent to ω. Let the circumcircle of $\triangle ABC$ intersect the circumcircle of $\triangle AXY$ at T.

Prove that $\angle BTI = \angle YTI$.